

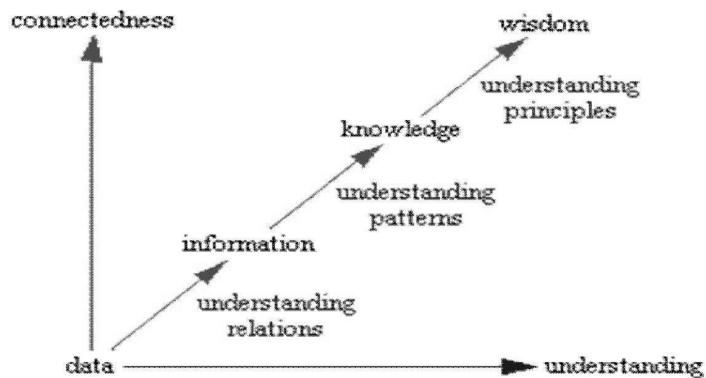
MGT 3213 - Management Information Systems

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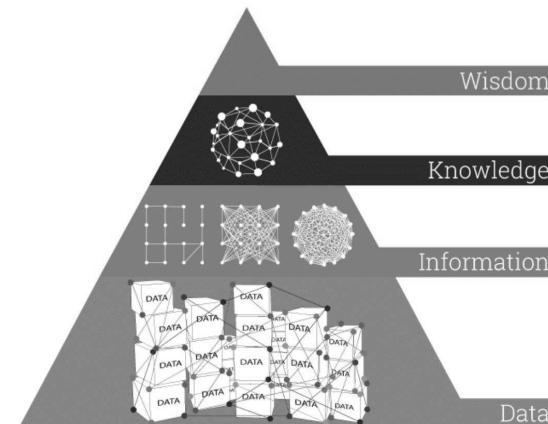
Content

- Management Information Systems
- Information Systems (IS) and Uses
- Gaining Competitive Advantage with Information Systems
- The Strategic Role of Information Systems
- Information Systems Planning
- Information Systems Development
- The Database Approach to Data Management
- Data Communications and Networking

From Data to Information to Knowledge and Wisdom

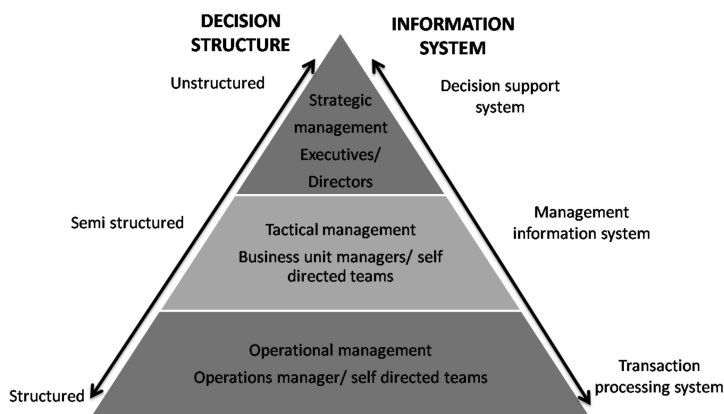


What is the DIKW Pyramid?



Each step up the pyramid answers questions about and adds value to the initial data.

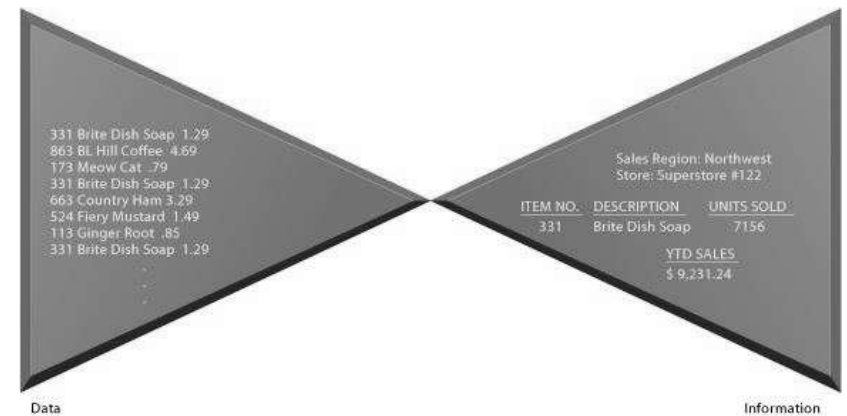
Information System for strategic advantage



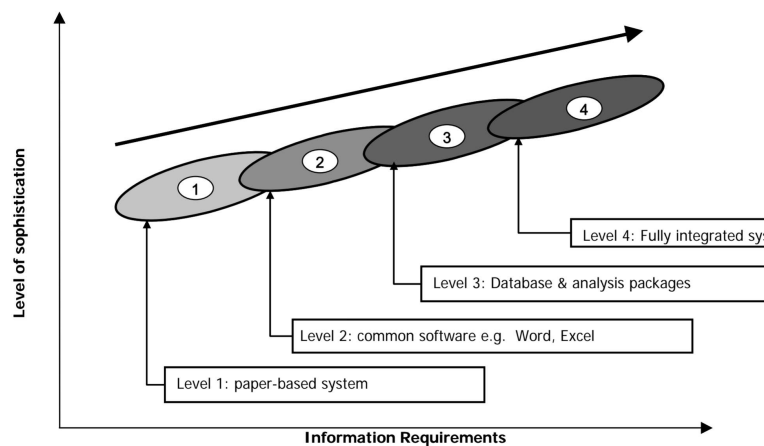
Information Category

- Strategic Information (Unstructured)
- Tactical Information (Semi-structured)
- Operational Information (Structured)

Data Vs. Information



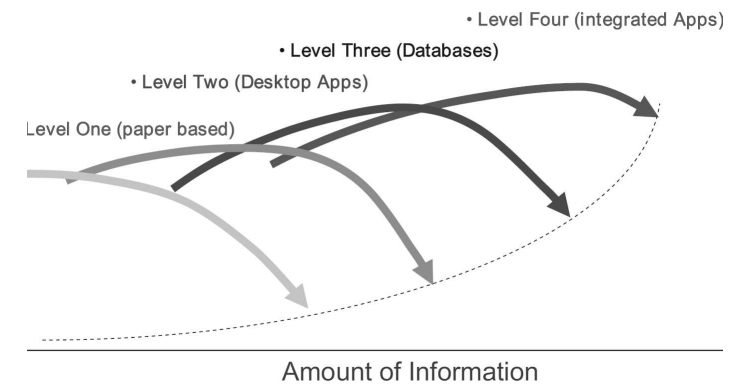
Sophistication



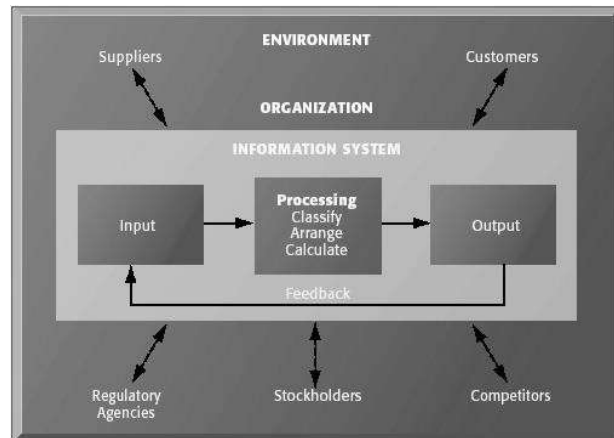
- These levels help to define the technology required based on the information requirements of a project or work:

Efficiency of MIS

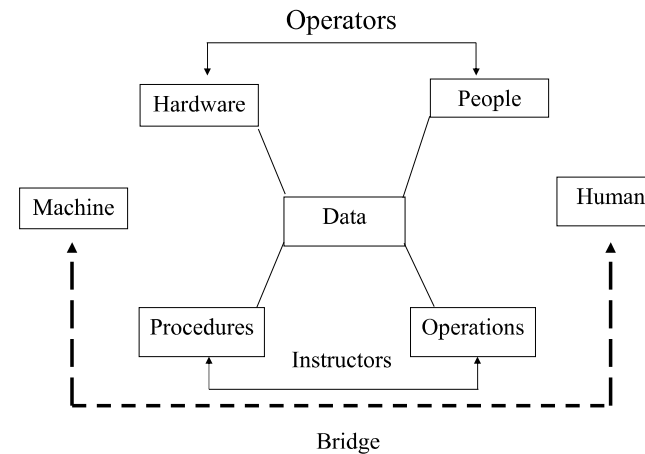
- The graph below shows the relationship between the volume of information and the quality of an information system comparing the four different levels shown above
- Each level drops in its efficiency as the volume of information grows. Managers need to identify when to move to the next level of IS that I needed for their organisation



Functions of an information system



Information Systems



- An information system (IS) can be any organized combination of:
 - people, hardware, software, communications networks, data resources, and policies and procedures
 - that collect, organise, stores, retrieves, transforms, and disseminates information in an organization.

Information Systems In Organisations

- It combines Business with Computer Science to complete the three major roles of an Information System:
 1. Support business processes and operations;
 2. support business decision making; and
 3. support competitive advantage.

Different kinds of Information Systems

- Transaction Processing Support (TPS)
- Office Automation Systems (OAS)
- The Management Information System (MIS)
- Executive information system (EIS)
- Decision Support System (DSS)
- Expert Systems (ES)
- Work Group Support System (WGSS)
- Financial Information System (FIS)
- Project Management Information System (PIS)



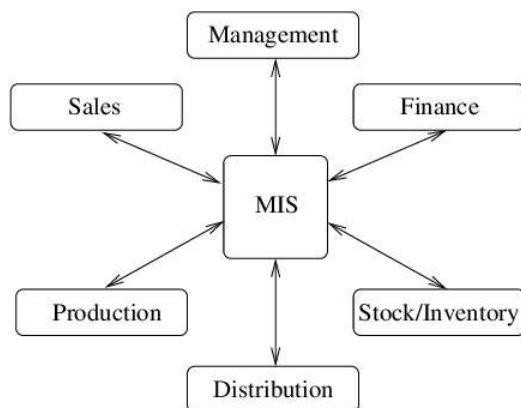
Definition of MIS

- A Management Information System (MIS) is an organized system for:
 - collecting,
 - storing,
 - processing, and
 - delivering information
- to support management operations and decision-making in an organization.

Functions of MIS

- **Data Collection:**
 - Gathering data from different sources (sales, production, HR, finance, etc.).
- **Data Processing:**
 - Converting raw data into meaningful information.
- **Information Storage:**
 - Keeping data in databases for future use.
- **Information Dissemination:**
 - Distributing reports and summaries to managers.
- **Decision Support:**
 - Helping managers make informed decisions using reports and analytics.

Management Information System



- MIS divided up into typical business operations or functions for support.
- **Sales MIS:**
 - Tracks sales performance by region, product, or salesperson.
- **Financial MIS:**
 - Provides budget reports, profit analysis, and financial statements.
- **Human Resource MIS:**
 - Manages employee records, payroll, and performance data.
- **Stock/Inventory MIS:**
 - Monitors stock levels, reorder points, and supplier performance.

Benefits of MIS

- Improves **decision-making quality**.
- Increases **efficiency and productivity**.
- Enhances **communication and coordination**.
- Provides **real-time and accurate information**.
- Helps in **strategic planning and forecasting**.

Example in Practice:

- A supermarket chain uses an MIS to track daily sales data from all its branches.
 - The system summarizes this data into weekly and monthly sales reports.
 - Managers use these reports to identify which products sell best and adjust inventory levels or marketing campaigns accordingly.

Thank you